

Kubernetes (K8s)

Kubernetes is an open source container orchestration engine for automating deployment, scaling, and management of containerized applications. The open source project is hosted by the Cloud Native Computing Foundation (CNCF).

It provides a scalable and resilient framework for automating the deployment, scaling, and management of applications across clusters of servers.

A SMALL HISTORY OF K8S:

☐ In the early 2000s, Google started developing a system called Borg to manage their internal containerized applications.

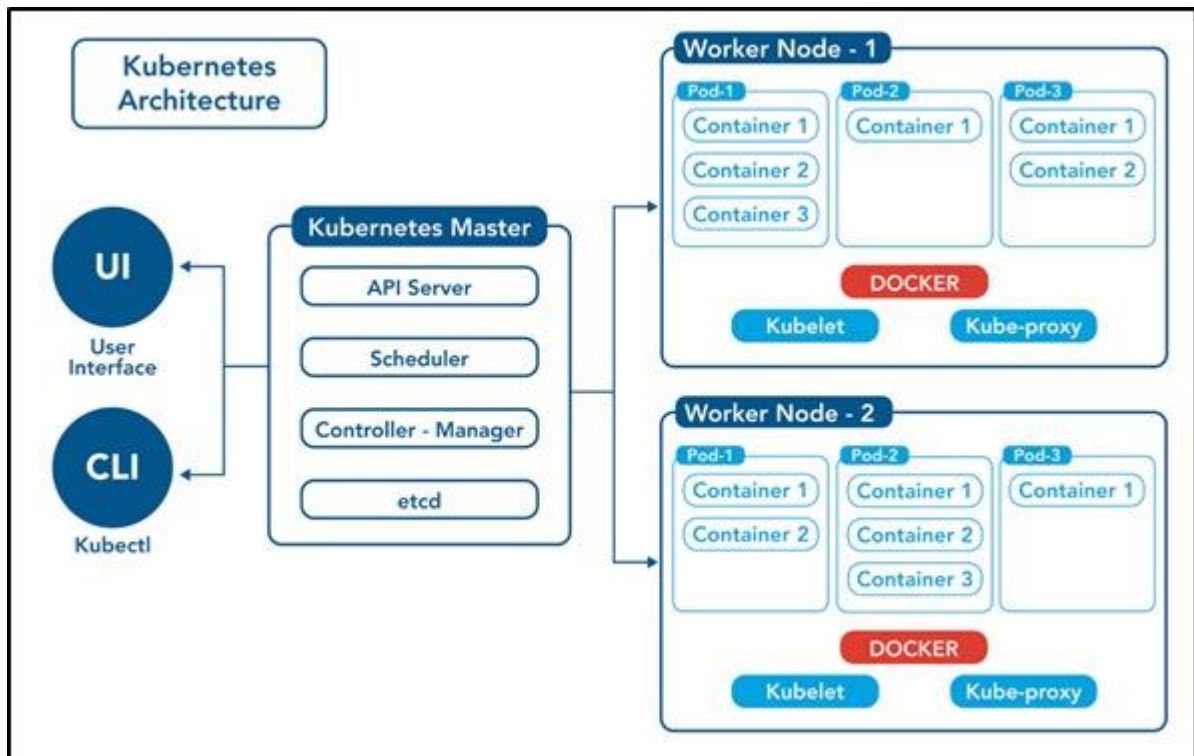
☐ Borg enabled Google to run applications at scale, providing features such as automatic scaling, service discovery, and fault tolerance.

☐ In 2014, Google open-sourced a version of Borg called Kubernetes.

☐ Kubernetes was donated to the Cloud Native Computing Foundation (CNCF), a neutral home for open-source cloud-native projects, in July 2015.

☐ Kubernetes 1.8 added significant enhancements for storage, security, and networking. Key features included the stable release of the stateful sets API, expanded support for volume plugins, and improvements in security policies.

☐ Check URL: <https://kubernetes.io/releases/> for more release details.



Control Plane /Master Node

The control plane's components make global decisions about the cluster (for example, scheduling), as well as detecting and responding to cluster events (for example, starting up a new pod when a deployment's replicas field is unsatisfied).

Control plane components can be run on any machine in the cluster. Do not run user containers on this machine.

Node Components / Worker Nodes

Node components run on every node, maintaining running pods and providing the Kubernetes runtime environment.

1. **Master Node:** The master node is responsible for managing the cluster and coordinating the overall state of the system. It includes the following components:
 - a. **API Server:** The API server is the central control point for all interactions with the cluster. It exposes the Kubernetes API and handles requests from users and other components.
 - b. **Scheduler:** The scheduler is responsible for assigning workloads (pods) to individual worker nodes based on resource requirements, constraints, and other policies.

c. Controller Manager: The controller manager runs various controllers that monitor the cluster state and drive it towards the desired state. Examples include the replication controller, node controller, and service controller.

d. etcd: etcd is a distributed key-value store used by Kubernetes to store cluster state and configuration data.

Services (short name = svc):

Service is an abstraction that defines a logical set of pods and a policy to access them. Services enable network connectivity and load balancing to the pods that are part of the service, allowing other components within or outside the cluster to interact with the application.

Service Types: Kubernetes supports different types of services:

1. NodePort: Exposes the service on a static port on each selected node's IP. This type makes the service accessible from outside the cluster by the <NodeIP>:<NodePort> combination.
2. ClusterIP: Exposes the service on a cluster-internal IP. This type makes the service only reachable within the cluster.
3. LoadBalancer: Creates an external load balancer in cloud environments, which routes traffic to the service.

Pod: The basic building block of Kubernetes. A pod represents a single instance of a running process within the cluster. It can encapsulate one or more containers that share the same network and storage resources.

1. Create a pod using run command

```
$ kubectl run <pod-name> --image=<image-name> --port=<container-port>
```

```
$ kubectl run my-pod --image=nginx --port=80
```

2. View all the pods

(In default namespace)

```
$ kubectl get pods
```

(In All namespace)

```
$ kubectl get pods -A
```

For a specific namespace

```
$ kubectl get pods -n kube-system
```

For a specific type

```
$ kubectl get pods <pod-name>
```

```
$ kubectl get pods <pod-name> -o wide
```

```
$ kubectl get pods <pod-name> -o yaml
```

```
$ kubectl get pods <pod-name> -o json
```

3. Describe a pod (View Pod details)

```
$ kubectl describe pod <pod-name>
```

```
$ kubectl describe pod my-pod
```

4. View Logs of a pod

```
$ kubectl logs <pod-name>
```

```
$ kubectl logs my-pod
```

5. Execute any command inside Pod (Inside Pod OS)

```
$ kubectl exec <pod-name> -- <command>
```

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  name: my-pod
```

labels:

app: my-web-app

type: backend

spec:

containers:

- name: nginx-container

image: nginx

ports:

- containerPort: 80

apiVersion: v1

kind: Pod

metadata:

name: my-app

spec:

containers:

- name: my-app-container

image: <images> ports:

- containerPort: 9090

kind: ReplicaSet

metadata:

name: my-rs

labels:

name: my-rs

spec:

replicas: 4

selector:

matchLabels:

apptype: web-backend

```
template:
  metadata:
    labels:
      apptype: web-backend
  spec:
    containers:
      - name: my-app
        image:
        ports:
          - containerPort: 8080
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: my-deploy
```

```
  labels:
```

```
    name: my-deploy
```

```
spec:
```

```
  replicas: 4
```

```
  selector:
```

```
    matchLabels:
```

```
      apptype: web-backend
```

```
  strategy:
```

```
    type: RollingUpdate
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        apptype: web-backend
```

```
    spec:
```

```
  containers:
  - name: my-app
    image:
    ports:
      - containerPort: 7070
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-deploy
  labels:
    name: my-deploy
spec:
  replicas: 1
  selector:
    matchLabels:
      apptype: web-backend
  strategy:
    type: RollingUpdate
  template:
    metadata:
      labels:
        apptype: web-backend
    spec:
      containers:
      - name: my-app
        image:
        ports:
          - containerPort: 7070
```

apiVersion: v1

kind: Service

metadata:

name: my-service

labels:

app: my-service

type: backend-app

spec:

type: NodePort

ports:

- targetPort: 7070

port: 7070

nodePort: 30002

selector:

apptype: web-backend

apiVersion: apps/v1

kind: Deployment

metadata:

name: my-deploy

labels:

name: my-deploy

spec:

replicas: 1

selector:

matchLabels:

apptype: web-backend

```
strategy:
  type: RollingUpdate
template:
  metadata:
    labels:
      apptype: web-backend
  spec:
    containers:
      - name: my-app
        image:
        ports:
          - containerPort: 9000
```

```
apiVersion: v1
kind: Service
metadata:
  name: my-service
  labels:
    app: my-service
spec:
  type: NodePort
  ports:
    - port: 9000
      targetPort: 8080
      nodePort: 30002
  selector:
```

apptype: web-backend

Namespace (short name = ns):

namespace is a virtual cluster or logical partition within a cluster that provides a way to organize and isolate resources. It allows multiple teams or projects to share the same physical cluster while maintaining resource separation and access control.

To create a namespace:

```
$ kubectl create namespace <namespace-name>
```

```
$ kubectl create ns my-bank
```

To switch to a specific namespace: (make this as default type)

```
$ kubectl config set-context --current --namespace=<namespace-name>
```

To list all namespaces:

```
$ kubectl get namespaces
```

To get resources within a specific namespace:

```
$ kubectl get <resource-type> -n <namespace-name>
```

```
$ kubectl get deploy -n my-bank
```

```
$ kubectl get deploy --namespace my-bank
```

```
$ kubectl get all --namespace my-bank
```

To delete a namespace and all associated resources:

```
$ kubectl delete namespace <namespace-name>
```

```
$ kubectl delete ns my-bank
```

```
kubectl create ns mydeploy
```

```
kubectl apply -f deploy.yml -n mydeploy
```

Create Deployment by executing above YAML file

```
$ kubectl create -f web-deploy.yml
```

Do necessary modifications if exist, else create new

```
$ kubectl create -f web-deploy.yml
```

Completely Modify Pod Template

```
$ kubectl replace -f web-deploy.yml
```

3. View Deployments

```
$ kubectl get deployments
```

```
$ kubectl get deploy
```

```
$ kubectl get deploy -o wide
```

```
$ kubectl get deploy <deployment-name> -o json
```

```
$ kubectl get deploy <deployment-name> -o yaml
```

4. View Deployment Description

```
$ kubectl describe deploy <deployment-name>
```

5. We can modify generated/updated YAML file

```
$ kubectl edit deploy <deployment-name>
```

```
## change replicas: count to any other value then (ESC):wq
```

```
# We can modify our YAML file and then execute apply command
```

```
$ kubectl apply -f web-deploy.yml
```

```
## We can Even scale using command also
```

```
$ kubectl scale deploy <deployment-name> --replicas=<desired-replica-count>
```

6. Delete Deployment

```
$ kubectl delete deploy <deployment-name>
```

```
$ kubectl delete -f web-deploy.yml
```

```
Create ReplicaSet by executing above YAML file
```

```
$ kubectl create -f rs-test.yml
```

```
# Do necessary modifications if exist, else create new
```

```
$ kubectl apply -f rs-test.yml
```

Completely Modify Pod Template

```
$ kubectl replace -f rs-test.yml
```

3. View ReplicaSets

```
$ kubectl get replicaset
```

```
$ kubectl get rs
```

```
$ kubectl get rs -o wide
```

```
$ kubectl get rs <replica-set-name> -o json
```

```
$ kubectl get rs <replica-set-name> -o yaml
```

4. View ReplicaSet Description

```
$ kubectl describe rs <replica-set-name>
```

5. We can modify generated/updated YAML file

```
$ kubectl edit rs <replica-set-name>
```

change replicas: count to any other value then (ESC):wq

We can modify our YAML file and then execute apply command

```
$ kubectl apply -f rs-test.yml
```

We can Even scale using command also

```
$ kubectl scale replicaset <replicaset-name> --replicas=<desired-replica-count>
```

6. Delete ReplicaSet

```
$ kubectl delete rs <replica-set-name>
```

```
$ kubectl delete -f rs-test.yml
```

```
Normal ScalingReplicaSet 26s deployment-controller Scaled up replica set my-deployment-f47778678 from 0 to 3
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
arvind_2004@ARVINDM:~/my_project$ kubectl create deployment webnginx2 --image=nginx:latest --replicas=1
deployment.apps/webnginx2 created
arvind_2004@ARVINDM:~/my_project$ kubectl replace -f deployment.yml
deployment.apps/my-deployment replaced
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME                                READY   STATUS             RESTARTS   AGE
my-deployment-f47778678-5qspr       0/1     ContainerCreating   0           12s
my-deployment-f47778678-c8d5m       0/1     ErrImagePull        0           12s
my-deployment-f47778678-czpqr       0/1     ImagePullBackOff    0           4m32s
my-deployment-f47778678-d5bxg       0/1     ImagePullBackOff    0           4m32s
my-deployment-f47778678-d8f7j       0/1     ImagePullBackOff    0           4m32s
my-deployment-f47778678-f82gp       0/1     ContainerCreating   0           12s
my-deployment-f47778678-fml6f       0/1     ErrImagePull        0           12s
my-deployment-f47778678-zrh84       0/1     ContainerCreating   0           12s
webnginx2-85b94f8d56-gsz4q          1/1     Running             0           72s
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy my-deploy --replicas=2
error: no objects passed to scale deployments.apps "my-deploy" not found
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy mydeploy --replicas=2
error: no objects passed to scale deployments.apps "mydeploy" not found
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy my-deploy --replicas=2
error: no objects passed to scale deployments.apps "my-deploy" not found
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy my-deployment --replicas=2
deployment.apps/my-deployment scaled
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME                                READY   STATUS             RESTARTS   AGE
my-deployment-f47778678-czpqr       0/1     ImagePullBackOff    0           10m
my-deployment-f47778678-d8f7j       0/1     ImagePullBackOff    0           10m
webnginx2-85b94f8d56-gsz4q          1/1     Running             0           7m13s
arvind_2004@ARVINDM:~/my_project$ kubectl delete webnginx2-85b94f8d56-gsz4q
error: the server doesn't have a resource type "webnginx2-85b94f8d56-gsz4q"
arvind_2004@ARVINDM:~/my_project$ kubectl delete pod webnginx2-85b94f8d56-gsz4q

mypod 1/1 Running 0 82m
arvind_2004@ARVINDM:~/my_project$ sudo nano rs-test.yml
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f rs-test.yml
error: error validating "rs-test.yml": error validating data: apiVersion not set; if you choose to ignore these errors, turn validation off with --validate=false
arvind_2004@ARVINDM:~/my_project$ sudo nano rs-test.yml
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f rs-test.yml
error: error validating "rs-test.yml": error validating data: apiVersion not set; if you choose to ignore these errors, turn validation off with --validate=false
arvind_2004@ARVINDM:~/my_project$ sudo nano rs-test.yml
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f rs-test.yml
replicaset.apps/my-rs created
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
NAME    READY   STATUS             RESTARTS   AGE
my-app1 0/1     ImagePullBackOff    0           11m
my-pod   1/1     Running             0           90m
my-pod2  1/1     Running             0           89m
my-rs-7xjf6 0/1     ContainerCreating   0           8s
my-rs-fd6z6 0/1     ContainerCreating   0           8s
my-rs-h8klx 0/1     ErrImagePull        0           8s
my-rs-wcqzw 0/1     ContainerCreating   0           8s
mypod    1/1     Running             0           93m
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME    READY   STATUS             RESTARTS   AGE
my-app1 0/1     ImagePullBackOff    0           11m
my-pod   1/1     Running             0           90m
my-pod2  1/1     Running             0           89m
my-rs-7xjf6 0/1     ErrImagePull        0           11s
my-rs-fd6z6 0/1     ContainerCreating   0           11s
my-rs-h8klx 0/1     ErrImagePull        0           11s
my-rs-wcqzw 0/1     ContainerCreating   0           11s
mypod    1/1     Running             0           93m
arvind_2004@ARVINDM:~/my_project$ kubectl delete pod my-rs-7xjf6
```

```
arvind_2004@ARVINDM: ~/m x + v
4b1990a..a515df1 day4 -> day4
arvind_2004@ARVINDM:~/my_project$ sudo nano my-pod.yml
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME      READY   STATUS    RESTARTS   AGE
my-pod    1/1     Running   0           76m
my-pod2   1/1     Running   0           75m
mypod     1/1     Running   0           79m
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
my-pod    1/1     Running   0           76m
my-pod2   1/1     Running   0           75m
mypod     1/1     Running   0           79m
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
my-pod    1/1     Running   0           77m
my-pod2   1/1     Running   0           76m
mypod     1/1     Running   0           80m
arvind_2004@ARVINDM:~/my_project$ pod run scale
Command 'pod' not found, did you mean:
  command 'od' from deb coreutils (9.4-2ubuntu2)
  command 'pdd' from deb pdd (1.7-1)
  command 'pom' from deb bsdgames (2.17-30)
  command 'pd' from deb puredata-core (0.54.1+ds-3)
  command 'wod' from deb sope-bin (5.9.0-1)
  command 'god' from deb ruby-god (0.13.7-3build5)
  command 'psd' from deb profile-sync-daemon (6.50-1)
  command 'poa' from deb poa (2.0+20060928-8)
  command 'pox' from deb python3-pox (0.3.3-1)
  command 'pwd' from deb coreutils (9.4-2ubuntu2)
  command 'pods' from deb pmtools (2.2.0-3)
  command 'pon' from deb ppp (2.4.9-1+1.1ubuntu1)
  command 'mod' from deb monodoc-base (6.8.0.105+dfsg-3.5ubuntu1)
Try: sudo apt install <deb name>

arvind_2004@ARVINDM: ~/m x + v
/.git: Permission denied
arvind_2004@ARVINDM:/$ kubectl exec -it my-pod2 -- /bin/bash
root@my-pod2:/# ls
bin  dev                docker-entrypoint.sh  home  lib64  mnt  proc  run  srv  tmp  var
boot  docker-entrypoint.d  etc                  lib   media  opt  root  sbin  sys  usr
root@my-pod2:/# exit
exit
arvind_2004@ARVINDM:/$ git init
/.git: Permission denied
arvind_2004@ARVINDM:/$ cd ~/my_project
-bash: cd: /home/arvind_2004/my_project: No such file or directory
arvind_2004@ARVINDM:/$ cd ~
arvind_2004@ARVINDM:~$ mkdir my_project
arvind_2004@ARVINDM:~$ cd my_project
arvind_2004@ARVINDM:~/my_project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/arvind_2004/my_project/.git/
arvind_2004@ARVINDM:~/my_project$ git add pom1.yml
fatal: pathspec 'pom1.yml' did not match any files
arvind_2004@ARVINDM:~/my_project$ ls -l pom1.yml
ls: cannot access 'pom1.yml': No such file or directory
arvind_2004@ARVINDM:~/my_project$ cd /
arvind_2004@ARVINDM:/$ cat pom1.yml
cat: pom1.yml: No such file or directory
```

```
arvind_2004@ARVINDM: ~/m x + v
}
arvind_2004@ARVINDM:/$ sudo nano pod.yml
[sudo] password for arvind_2004:
arvind_2004@ARVINDM:/$ kubectl logs my-pod2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/03/20 04:46:47 [notice] 1#1: using the "epoll" event method
2025/03/20 04:46:47 [notice] 1#1: nginx/1.27.4
2025/03/20 04:46:47 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)
2025/03/20 04:46:47 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
2025/03/20 04:46:47 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/03/20 04:46:47 [notice] 1#1: start worker processes
2025/03/20 04:46:47 [notice] 1#1: start worker process 29
2025/03/20 04:46:47 [notice] 1#1: start worker process 30
2025/03/20 04:46:47 [notice] 1#1: start worker process 31
2025/03/20 04:46:47 [notice] 1#1: start worker process 32
2025/03/20 04:46:47 [notice] 1#1: start worker process 33
2025/03/20 04:46:47 [notice] 1#1: start worker process 34
2025/03/20 04:46:47 [notice] 1#1: start worker process 35
2025/03/20 04:46:47 [notice] 1#1: start worker process 36
2025/03/20 04:46:47 [notice] 1#1: start worker process 37
2025/03/20 04:46:47 [notice] 1#1: start worker process 38
2025/03/20 04:46:47 [notice] 1#1: start worker process 39
2025/03/20 04:46:47 [notice] 1#1: start worker process 40
arvind_2004@ARVINDM:/$ kubectl get pod

arvind_2004@ARVINDM: ~/m x + v
arvind_2004@ARVINDM:/$ ls
bin          boot  etc  init  lib.usr-is-merged  lost+found  mnt  proc  run  sbin.usr-is-merged  srv  tmp  var
bin.usr-is-merged  dev  home  lib  lib64  media  opt  root  sbin  snap
arvind_2004@ARVINDM:/$ kubectl get pod -o wide
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE       NOMINATED NODE   READINESS GATES
my-pod    1/1     Running   0           3m37s  10.244.0.6    minikube   <none>            <none>
my-pod2   1/1     Running   0           2m38s  10.244.0.7    minikube   <none>            <none>
mypod     1/1     Running   0           6m32s  10.244.0.5    minikube   <none>            <none>
arvind_2004@ARVINDM:/$ kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
coredns-668d6bf9bc-gj2j2            1/1     Running   1 (58m ago)  20h
etcd-minikube                       1/1     Running   1 (58m ago)  20h
kube-apiserver-minikube             1/1     Running   1 (58m ago)  20h
kube-controller-manager-minikube    1/1     Running   1 (58m ago)  20h
kube-proxy-2chv9                    1/1     Running   1 (58m ago)  20h
kube-scheduler-minikube             1/1     Running   1 (58m ago)  20h
storage-provisioner                 1/1     Running   3 (58m ago)  20h
arvind_2004@ARVINDM:/$ kubectl get pods my-pod2 -o yaml
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: "2025-03-20T04:46:43Z"
  labels:
    run: my-pod2
  name: my-pod2
  namespace: default
  resourceVersion: "22354"
  uid: c2ee31e-a019-40aa-b8fb-b5f3b9ddfd3f
spec:
  containers:
  - image: nginx:latest
    imagePullPolicy: Always
    name: my-pod2
```

```
arvind_2004@ARVINDM: ~/m x + v
arvind_2004@ARVINDM:/$ kubectl run my-pod2 --image=nginx:latest --port=8084
pod/my-pod2 created
arvind_2004@ARVINDM:/$ kubectl get pod
NAME      READY   STATUS    RESTARTS   AGE
my-pod    1/1     Running   0           62s
my-pod2   0/1     ContainerCreating   0           3s
mypod     1/1     Running   0           3m57s
arvind_2004@ARVINDM:/$ kubectl logs my-pod2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/03/20 04:46:47 [notice] 1#1: using the "epoll" event method
2025/03/20 04:46:47 [notice] 1#1: nginx/1.27.4
2025/03/20 04:46:47 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)
2025/03/20 04:46:47 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
2025/03/20 04:46:47 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/03/20 04:46:47 [notice] 1#1: start worker processes
2025/03/20 04:46:47 [notice] 1#1: start worker process 29
2025/03/20 04:46:47 [notice] 1#1: start worker process 30
2025/03/20 04:46:47 [notice] 1#1: start worker process 31
2025/03/20 04:46:47 [notice] 1#1: start worker process 32
2025/03/20 04:46:47 [notice] 1#1: start worker process 33
2025/03/20 04:46:47 [notice] 1#1: start worker process 34
2025/03/20 04:46:47 [notice] 1#1: start worker process 35
2025/03/20 04:46:47 [notice] 1#1: start worker process 36
2025/03/20 04:46:47 [notice] 1#1: start worker process 37
2025/03/20 04:46:47 [notice] 1#1: start worker process 38

arvind_2004@ARVINDM: ~/m x + v
arvind_2004@ARVINDM:/$ minikube start
🔧 minikube v1.35.0 on Ubuntu 24.04 (amd64)
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.46 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.32.0 on Docker 27.4.1 ...
🔍 Verifying Kubernetes components...
  • Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: default-storageclass, storage-provisioner
👉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
arvind_2004@ARVINDM:/$ kubectl get pod
No resources found in default namespace.
arvind_2004@ARVINDM:/$ kubectl run mypod --image=nginx --port=80
pod/mypod created
arvind_2004@ARVINDM:/$ kubectl get pod
NAME      READY   STATUS    RESTARTS   AGE
mypod     1/1     Running   0           73s
arvind_2004@ARVINDM:/$ kubectl get node
NAME      STATUS   ROLES    AGE     VERSION
minikube  Ready    control-plane  19h    v1.32.0
arvind_2004@ARVINDM:/$ kubectl delete pod my-pod
Error from server (NotFound): pods "my-pod" not found
arvind_2004@ARVINDM:/$ kubectl delete pod mypod
pod "mypod" deleted
arvind_2004@ARVINDM:/$ kubectl run mypod --image=nginx --port=8085
pod/mypod created
arvind_2004@ARVINDM:/$ kubectl get pod
NAME      READY   STATUS    RESTARTS   AGE
mypod     1/1     Running   0           12s
arvind_2004@ARVINDM:/$ kubectl run mypod --image=nginx:latest --port=8087
Error from server (AlreadyExists): pods "mypod" already exists
arvind_2004@ARVINDM:/$ kubectl run my-pod --image=nginx:latest --port=8087
```

```
arvind_2004@ARVINDM: ~/m x + v
arvind_2004@ARVINDM:~/my_project$ kubectl delete all --all
pod "my-deploy-6b478cb7cb-4fhwv" deleted
service "kubernetes" deleted
service "my-service" deleted
deployment.apps "my-deploy" deleted
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
deployment.apps/my-deploy created
service/my-service created
arvind_2004@ARVINDM:~/my_project$ minikube service my-service
+-----+-----+-----+-----+
| NAMESPACE | NAME   | TARGET PORT | URL                               |
+-----+-----+-----+-----+
| default    | my-service | 9001         | http://192.168.49.2:30005       |
+-----+-----+-----+-----+
* Starting tunnel for service my-service.
+-----+-----+-----+-----+
| NAMESPACE | NAME   | TARGET PORT | URL                               |
+-----+-----+-----+-----+
| default    | my-service |             | http://127.0.0.1:36227         |
+-----+-----+-----+-----+
🔗 Opening service default/my-service in default browser...
🔗 http://127.0.0.1:36227
! Because you are using a Docker driver on linux, the terminal needs to be open to run it.
^C Stopping tunnel for service my-service.
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005
<!doctype html><html lang="en"><head><title>HTTP Status 404 - Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,
sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {
font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 -
Not Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b> The origin server did not find a current repre
sentation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /><h3>Apache Tomcat/9.0.102</h3>
</body></html>arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-app/
<!doctype html><html lang="en"><head><title>HTTP Status 404 - Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,
sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {
font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 -
Not Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b> The origin server did not find a current repre
sentation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /><h3>Apache Tomcat/9.0.102</h3>
</body></html>arvind_2004@ARVINDM:~/my_project$ kubectl exec -it my-deploy-6b478cb7cb-t5lkg -- /bin/bash
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat# ls
bin          conf          filtered-KEYS  LICENSE      native-jni-lib  README.md     RUNNING.txt   upstream-KEYS  webapps.dist
BUILDING.txt CONTRIBUTING.md lib            Logs         NOTICE         RELEASE-NOTES temp          webapps       work
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat# cd webapps
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat/webapps# ls
my-web  my-web.war
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat/webapps# cd
root@my-deploy-6b478cb7cb-t5lkg:~# exit
exit
arvind_2004@ARVINDM:~/my_project$ ls
deployment.yml kubectl.txt my-pod.yml rs-test.yml
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-web/
<html>
<body>
<h2>Hello World!</h2>
</body>
</html>
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-web/
<html>
<body>
<h2>Hello World!</h2>
</body>
</html>
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-web/
```

Microsoft Windows [Version 10.0.22631.4751]

(c) Microsoft Corporation. All rights reserved.

C:\Users\Arvind M>wsl.exe -d Ubuntu

Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

* Documentation: <https://help.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/pro>

System information as of Thu Mar 20 04:31:21 UTC 2025

System load: 0.83 Processes: 36

Usage of /: 0.8% of 1006.85GB Users logged in: 0

Memory usage: 12% IPv4 address for eth0: 172.27.90.197

Swap usage: 0%

This message is shown once a day. To disable it please create the
/home/arvind_2004/.hushlogin file.

arvind_2004@ARVINDM:/mnt/c/Users/Arvind M\$ cd ..

arvind_2004@ARVINDM:/mnt/c/Users\$ cd /

arvind_2004@ARVINDM:/\$ minikube start

😊 minikube v1.35.0 on Ubuntu 24.04 (amd64)

🔧 Using the docker driver based on existing profile

👍 Starting "minikube" primary control-plane node in "minikube" cluster

🚚 Pulling base image v0.0.46 ...

🔄 Restarting existing docker container for "minikube" ...

👉 Preparing Kubernetes v1.32.0 on Docker 27.4.1 ...

🔍 Verifying Kubernetes components...

- Using image gcr.io/k8s-minikube/storage-provisioner:v5

✨ Enabled addons: default-storageclass, storage-provisioner

🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

arvind_2004@ARVINDM:/\$ kubectl get pod

No resources found in default namespace.

```
arvind_2004@ARVINDM:/$ kubectl run mypod --image=nginx --port=80
```

pod/mypod created

```
arvind_2004@ARVINDM:/$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

mypod	1/1	Running	0	73s
-------	-----	---------	---	-----

```
arvind_2004@ARVINDM:/$ kubectl get node
```

NAME	STATUS	ROLES	AGE	VERSION
------	--------	-------	-----	---------

minikube	Ready	control-plane	19h	v1.32.0
----------	-------	---------------	-----	---------

```
arvind_2004@ARVINDM:/$ kubectl delete pod my-pod
```

Error from server (NotFound): pods "my-pod" not found

```
arvind_2004@ARVINDM:/$ kubectl delete pod mypod
```

pod "mypod" deleted

```
arvind_2004@ARVINDM:/$ kubectl run mypod --image=nginx --port=8085
```

pod/mypod created

```
arvind_2004@ARVINDM:/$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

mypod	1/1	Running	0	12s
-------	-----	---------	---	-----

```
arvind_2004@ARVINDM:/$ kubectl run mypod --image=nginx:latest --port=8087
```

Error from server (AlreadyExists): pods "mypod" already exists

```
arvind_2004@ARVINDM:/$ kubectl run my-pod --image=nginx:latest --port=8087
```

pod/my-pod created

```
arvind_2004@ARVINDM:/$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-pod	1/1	Running	0	8s
--------	-----	---------	---	----

mypod	1/1	Running	0	3m3s
-------	-----	---------	---	------

```
arvind_2004@ARVINDM:/$ kubectl run my-pod2 --image=nginx:latest --port=8084
```

pod/my-pod2 created

```
arvind_2004@ARVINDM:/$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
my-pod	1/1	Running	0	62s
my-pod2	0/1	ContainerCreating	0	3s
mypod	1/1	Running	0	3m57s

```
arvind_2004@ARVINDM:/$ kubectl logs my-pod2
```

```
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
```

```
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
```

```
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
```

```
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
```

```
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
```

```
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
```

```
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
```

```
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
```

```
/docker-entrypoint.sh: Configuration complete; ready for start up
```

```
2025/03/20 04:46:47 [notice] 1#1: using the "epoll" event method
```

```
2025/03/20 04:46:47 [notice] 1#1: nginx/1.27.4
```

```
2025/03/20 04:46:47 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)
```

```
2025/03/20 04:46:47 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
```

```
2025/03/20 04:46:47 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
```

```
2025/03/20 04:46:47 [notice] 1#1: start worker processes
```

```
2025/03/20 04:46:47 [notice] 1#1: start worker process 29
```

```
2025/03/20 04:46:47 [notice] 1#1: start worker process 30
```

```
2025/03/20 04:46:47 [notice] 1#1: start worker process 31
```

```
2025/03/20 04:46:47 [notice] 1#1: start worker process 32
```

```
2025/03/20 04:46:47 [notice] 1#1: start worker process 33
```

```
2025/03/20 04:46:47 [notice] 1#1: start worker process 34
```

2025/03/20 04:46:47 [notice] 1#1: start worker process 35

2025/03/20 04:46:47 [notice] 1#1: start worker process 36

2025/03/20 04:46:47 [notice] 1#1: start worker process 37

2025/03/20 04:46:47 [notice] 1#1: start worker process 38

2025/03/20 04:46:47 [notice] 1#1: start worker process 39

2025/03/20 04:46:47 [notice] 1#1: start worker process 40

arvind_2004@ARVINDM:/\$ minikube ssh

docker@minikube:~\$ docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
PORTS	NAMES			
2c3ef2e4ea91	nginx	"/docker-entrypoint...."	About a minute ago	Up About a minute
	k8s_my-pod2_my-pod2_default_c2eee31e-a019-40aa-b8fb-b5f3b9ddfd3f_0			
3c85de85a4ac	registry.k8s.io/pause:3.10	"/pause"	About a minute ago	Up About a minute
	k8s_POD_my-pod2_default_c2eee31e-a019-40aa-b8fb-b5f3b9ddfd3f_0			
18fefc6bc8ad	nginx	"/docker-entrypoint...."	2 minutes ago	Up 2 minutes
	k8s_my-pod_my-pod_default_c76700d2-2a03-4cd2-8887-9adc28ee0e72_0			
1961b345922d	registry.k8s.io/pause:3.10	"/pause"	2 minutes ago	Up 2 minutes
	k8s_POD_my-pod_default_c76700d2-2a03-4cd2-8887-9adc28ee0e72_0			
43ca4ffd4f14	nginx	"/docker-entrypoint...."	5 minutes ago	Up 5 minutes
	k8s_mypod_mypod_default_8671366b-5ac7-4c55-9a4a-40a746680ba7_0			
a81a4eab7452	registry.k8s.io/pause:3.10	"/pause"	5 minutes ago	Up 5 minutes
	k8s_POD_mypod_default_8671366b-5ac7-4c55-9a4a-40a746680ba7_0			
68039fa229f4	6e38f40d628d	"/storage-provisioner"	16 minutes ago	Up 16 minutes
	k8s_storage-provisioner_storage-provisioner_kube-system_662c7ba5-f633-49b7-8f08-8fe775a520ee_3			
c2f36b460128	c69fa2e9cbf5	"/coredns -conf /etc..."	16 minutes ago	Up 16 minutes
	k8s_coredns_coredns-668d6bf9bc-gj2j2_kube-system_7bfcff89-694f-406a-808e-466d0d549ea8_1			
20035d5be14e	040f9f8aac8c	"/usr/local/bin/kube..."	16 minutes ago	Up 16 minutes
	k8s_kube-proxy_kube-proxy-2chv9_kube-system_b55c713f-8f1e-4218-b286-07528e7c34ca_1			

```

45f42d177058 registry.k8s.io/pause:3.10 "/pause" 16 minutes ago Up 16
minutes      k8s_POD_coredns-668d6bf9bc-gj2j2_kube-system_7bfcff89-694f-406a-
808e-466d0d549ea8_1

1124f740e47b registry.k8s.io/pause:3.10 "/pause" 16 minutes ago Up 16
minutes      k8s_POD_storage-provisioner_kube-system_662c7ba5-f633-49b7-8f08-
8fe775a520ee_1

3bb28a2ca964 registry.k8s.io/pause:3.10 "/pause" 16 minutes ago Up 16
minutes      k8s_POD_kube-proxy-2chv9_kube-system_b55c713f-8f1e-4218-b286-
07528e7c34ca_1

13ad1eb82586 a9e7e6b294ba "etcd --advertise-cl..." 16 minutes ago Up 16
minutes      k8s_etcd_etcd-minikube_kube-
system_2b4b75c2a289008e0b381891e9683040_1

9c99aea7da36 c2e17b8d0f4a "kube-apiserver --ad..." 16 minutes ago Up 16
minutes      k8s_kube-apiserver_kube-apiserver-minikube_kube-
system_d72d0a4cf4be077c9919d46b7358a5e8_1

09bb410b3d82 8cab3d2a8bd0 "kube-controller-man..." 16 minutes ago Up 16
minutes      k8s_kube-controller-manager_kube-controller-manager-minikube_kube-
system_843c74f7b3bc7d7040a05c31708a6a30_1

cae7502e8a41 a389e107f4ff "kube-scheduler --au..." 16 minutes ago Up 16
minutes      k8s_kube-scheduler_kube-scheduler-minikube_kube-
system_d14ce008bee3a1f3bd7cf547688f9dfe_1

80138199261a registry.k8s.io/pause:3.10 "/pause" 16 minutes ago Up 16
minutes      k8s_POD_kube-apiserver-minikube_kube-
system_d72d0a4cf4be077c9919d46b7358a5e8_1

a0e780589b00 registry.k8s.io/pause:3.10 "/pause" 16 minutes ago Up 16
minutes      k8s_POD_etcd-minikube_kube-
system_2b4b75c2a289008e0b381891e9683040_1

4acf7db67313 registry.k8s.io/pause:3.10 "/pause" 16 minutes ago Up 16
minutes      k8s_POD_kube-controller-manager-minikube_kube-
system_843c74f7b3bc7d7040a05c31708a6a30_1

a00a5c570e77 registry.k8s.io/pause:3.10 "/pause" 16 minutes ago Up 16
minutes      k8s_POD_kube-scheduler-minikube_kube-
system_d14ce008bee3a1f3bd7cf547688f9dfe_1

```

```
docker@minikube:~$ exit
```

```
logout
```

```
arvind_2004@ARVINDM:/$ ls
```

```
bin          boot etc  init lib.usr-is-merged lost+found mnt proc run  sbin.usr-is-merged
srv tmp var
```

```
bin.usr-is-merged dev home lib lib64          media    opt root sbin snap          sys
usr
```

```
arvind_2004@ARVINDM:/$ kubectl get pod -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED	NODE
my-pod	1/1	Running	0	3m37s	10.244.0.6	minikube	<none>	<none>
my-pod2	1/1	Running	0	2m38s	10.244.0.7	minikube	<none>	<none>
mypod	1/1	Running	0	6m32s	10.244.0.5	minikube	<none>	<none>

```
arvind_2004@ARVINDM:/$ kubectl get pods -n kube-system
```

NAME	READY	STATUS	RESTARTS	AGE
coredns-668d6bf9bc-gj2j2	1/1	Running	1 (58m ago)	20h
etcd-minikube	1/1	Running	1 (58m ago)	20h
kube-apiserver-minikube	1/1	Running	1 (58m ago)	20h
kube-controller-manager-minikube	1/1	Running	1 (58m ago)	20h
kube-proxy-2chv9	1/1	Running	1 (58m ago)	20h
kube-scheduler-minikube	1/1	Running	1 (58m ago)	20h
storage-provisioner	1/1	Running	3 (58m ago)	20h

```
arvind_2004@ARVINDM:/$ kubectl get pods my-pod2 -o yaml
```

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  creationTimestamp: "2025-03-20T04:46:43Z"
```

```
  labels:
```

```
    run: my-pod2
```

```
  name: my-pod2
```

```
  namespace: default
```

resourceVersion: "22354"

uid: c2eee31e-a019-40aa-b8fb-b5f3b9ddfd3f

spec:

containers:

- image: nginx:latest

imagePullPolicy: Always

name: my-pod2

ports:

- containerPort: 8084

protocol: TCP

resources: {}

terminationMessagePath: /dev/termination-log

terminationMessagePolicy: File

volumeMounts:

- mountPath: /var/run/secrets/kubernetes.io/serviceaccount

name: kube-api-access-ff8h2

readOnly: true

dnsPolicy: ClusterFirst

enableServiceLinks: true

nodeName: minikube

preemptionPolicy: PreemptLowerPriority

priority: 0

restartPolicy: Always

schedulerName: default-scheduler

securityContext: {}

serviceAccount: default

serviceAccountName: default

terminationGracePeriodSeconds: 30

tolerations:

- effect: NoExecute

 - key: node.kubernetes.io/not-ready

 - operator: Exists

 - tolerationSeconds: 300

- effect: NoExecute

 - key: node.kubernetes.io/unreachable

 - operator: Exists

 - tolerationSeconds: 300

volumes:

- name: kube-api-access-ff8h2

 - projected:

 - defaultMode: 420

 - sources:

 - serviceAccountToken:

 - expirationSeconds: 3607

 - path: token

 - configMap:

 - items:

 - key: ca.crt

 - path: ca.crt

 - name: kube-root-ca.crt

 - downwardAPI:

 - items:

 - fieldRef:

 - apiVersion: v1

 - fieldPath: metadata.namespace

 - path: namespace

status:

conditions:

- lastProbeTime: null

lastTransitionTime: "2025-03-20T04:46:47Z"

status: "True"

type: PodReadyToStartContainers

- lastProbeTime: null

lastTransitionTime: "2025-03-20T04:46:43Z"

status: "True"

type: Initialized

- lastProbeTime: null

lastTransitionTime: "2025-03-20T04:46:47Z"

status: "True"

type: Ready

- lastProbeTime: null

lastTransitionTime: "2025-03-20T04:46:47Z"

status: "True"

type: ContainersReady

- lastProbeTime: null

lastTransitionTime: "2025-03-20T04:46:43Z"

status: "True"

type: PodScheduled

containerStatuses:

- containerID:

docker://2c3ef2e4ea91dddd6ada681eab9430a4888c1c17351df593c32dd7ada2f8c06b

image: nginx:latest

imageID: docker-

pullable://nginx@sha256:124b44bfc9ccd1f3cedf4b592d4d1e8bddb78b51ec2ed5056c52d3692baebc19

```
lastState: {}
name: my-pod2
ready: true
restartCount: 0
started: true
state:
  running:
    startedAt: "2025-03-20T04:46:47Z"
volumeMounts:
- mountPath: /var/run/secrets/kubernetes.io/serviceaccount
  name: kube-api-access-ff8h2
  readOnly: true
  recursiveReadOnly: Disabled
hostIP: 192.168.49.2
hostIPs:
- ip: 192.168.49.2
phase: Running
podIP: 10.244.0.7
podIPs:
- ip: 10.244.0.7
qosClass: BestEffort
startTime: "2025-03-20T04:46:43Z"
```

```
arvind_2004@ARVINDM:/$ kubectl get pods my-pod2 -o json
```

```
{
  "apiVersion": "v1",
  "kind": "Pod",
  "metadata": {
    "creationTimestamp": "2025-03-20T04:46:43Z",
```

```
"labels": {
  "run": "my-pod2"
},
"name": "my-pod2",
"namespace": "default",
"resourceVersion": "22354",
"uid": "c2eee31e-a019-40aa-b8fb-b5f3b9ddfd3f"
},
"spec": {
  "containers": [
    {
      "image": "nginx:latest",
      "imagePullPolicy": "Always",
      "name": "my-pod2",
      "ports": [
        {
          "containerPort": 8084,
          "protocol": "TCP"
        }
      ],
      "resources": {},
      "terminationMessagePath": "/dev/termination-log",
      "terminationMessagePolicy": "File",
      "volumeMounts": [
        {
          "mountPath": "/var/run/secrets/kubernetes.io/serviceaccount",
          "name": "kube-api-access-ff8h2",
          "readOnly": true
        }
      ]
    }
  ]
}
```

```
    }
  ]
}
],
"dnsPolicy": "ClusterFirst",
"enableServiceLinks": true,
"nodeName": "minikube",
"preemptionPolicy": "PreemptLowerPriority",
"priority": 0,
"restartPolicy": "Always",
"schedulerName": "default-scheduler",
"securityContext": {},
"serviceAccount": "default",
"serviceAccountName": "default",
"terminationGracePeriodSeconds": 30,
"tolerations": [
  {
    "effect": "NoExecute",
    "key": "node.kubernetes.io/not-ready",
    "operator": "Exists",
    "tolerationSeconds": 300
  },
  {
    "effect": "NoExecute",
    "key": "node.kubernetes.io/unreachable",
    "operator": "Exists",
    "tolerationSeconds": 300
  }
]
```

```
],  
"volumes": [  
  {  
    "name": "kube-api-access-ff8h2",  
    "projected": {  
      "defaultMode": 420,  
      "sources": [  
        {  
          "serviceAccountToken": {  
            "expirationSeconds": 3607,  
            "path": "token"  
          }  
        },  
        {  
          "configMap": {  
            "items": [  
              {  
                "key": "ca.crt",  
                "path": "ca.crt"  
              }  
            ],  
            "name": "kube-root-ca.crt"  
          }  
        },  
        {  
          "downwardAPI": {  
            "items": [  
              {
```

```
        "fieldRef": {
            "apiVersion": "v1",
            "fieldPath": "metadata.namespace"
        },
        "path": "namespace"
    }
]
}
}
]
}
}
]
},
"status": {
    "conditions": [
        {
            "lastProbeTime": null,
            "lastTransitionTime": "2025-03-20T04:46:47Z",
            "status": "True",
            "type": "PodReadyToStartContainers"
        },
        {
            "lastProbeTime": null,
            "lastTransitionTime": "2025-03-20T04:46:43Z",
            "status": "True",
            "type": "Initialized"
        },
    ],
}
```

```
{
  "lastProbeTime": null,
  "lastTransitionTime": "2025-03-20T04:46:47Z",
  "status": "True",
  "type": "Ready"
},
{
  "lastProbeTime": null,
  "lastTransitionTime": "2025-03-20T04:46:47Z",
  "status": "True",
  "type": "ContainersReady"
},
{
  "lastProbeTime": null,
  "lastTransitionTime": "2025-03-20T04:46:43Z",
  "status": "True",
  "type": "PodScheduled"
}
],
"containerStatuses": [
  {
    "containerID":
"docker://2c3ef2e4ea91dddd6ada681eab9430a4888c1c17351df593c32dd7ada2f8c06b",
    "image": "nginx:latest",
    "imageID": "docker-
pullable://nginx@sha256:124b44bfc9ccd1f3cedf4b592d4d1e8bddb78b51ec2ed5056c52d36
92baebc19",
    "lastState": {},
    "name": "my-pod2",
```

```
"ready": true,
"restartCount": 0,
"started": true,
"state": {
  "running": {
    "startedAt": "2025-03-20T04:46:47Z"
  }
},
"volumeMounts": [
  {
    "mountPath": "/var/run/secrets/kubernetes.io/serviceaccount",
    "name": "kube-api-access-ff8h2",
    "readOnly": true,
    "recursiveReadOnly": "Disabled"
  }
]
},
"hostIP": "192.168.49.2",
"hostIPs": [
  {
    "ip": "192.168.49.2"
  }
],
"phase": "Running",
"podIP": "10.244.0.7",
"podIPs": [
  {
```

```
        "ip": "10.244.0.7"
    }
],
    "qosClass": "BestEffort",
    "startTime": "2025-03-20T04:46:43Z"
}
}
```

arvind_2004@ARVINDM:/\$ sudo nano pod.yml

[sudo] password for arvind_2004:

arvind_2004@ARVINDM:/\$ kubectl logs my-pod2

/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration

/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/

/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh

10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf

10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf

/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh

/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh

/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh

/docker-entrypoint.sh: Configuration complete; ready for start up

2025/03/20 04:46:47 [notice] 1#1: using the "epoll" event method

2025/03/20 04:46:47 [notice] 1#1: nginx/1.27.4

2025/03/20 04:46:47 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)

2025/03/20 04:46:47 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2

2025/03/20 04:46:47 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576

2025/03/20 04:46:47 [notice] 1#1: start worker processes

2025/03/20 04:46:47 [notice] 1#1: start worker process 29

2025/03/20 04:46:47 [notice] 1#1: start worker process 30

2025/03/20 04:46:47 [notice] 1#1: start worker process 31
2025/03/20 04:46:47 [notice] 1#1: start worker process 32
2025/03/20 04:46:47 [notice] 1#1: start worker process 33
2025/03/20 04:46:47 [notice] 1#1: start worker process 34
2025/03/20 04:46:47 [notice] 1#1: start worker process 35
2025/03/20 04:46:47 [notice] 1#1: start worker process 36
2025/03/20 04:46:47 [notice] 1#1: start worker process 37
2025/03/20 04:46:47 [notice] 1#1: start worker process 38
2025/03/20 04:46:47 [notice] 1#1: start worker process 39
2025/03/20 04:46:47 [notice] 1#1: start worker process 40

arvind_2004@ARVINDM:/\$ kubectl get pod

NAME	READY	STATUS	RESTARTS	AGE
my-pod	1/1	Running	0	54m
my-pod2	1/1	Running	0	53m
mypod	1/1	Running	0	57m

arvind_2004@ARVINDM:/\$ sudo nano pod1.yml

arvind_2004@ARVINDM:/\$ kubectl get pod

NAME	READY	STATUS	RESTARTS	AGE
my-pod	1/1	Running	0	57m
my-pod2	1/1	Running	0	56m
mypod	1/1	Running	0	60m

arvind_2004@ARVINDM:/\$ kubectl exec -it my-pod2 -- /bin/bash

root@my-pod2:/# exit

exit

arvind_2004@ARVINDM:/\$ git init

/.git: Permission denied

arvind_2004@ARVINDM:/\$ kubectl exec -it my-pod2 -- /bin/bash

root@my-pod2:/# ls

bin dev docker-entrypoint.sh home lib64 mnt proc run srv tmp var

boot docker-entrypoint.d etc lib media opt root sbin sys usr

root@my-pod2:/# exit

exit

arvind_2004@ARVINDM:/\$ git init

/.git: Permission denied

arvind_2004@ARVINDM:/\$ cd ~/my_project

-bash: cd: /home/arvind_2004/my_project: No such file or directory

arvind_2004@ARVINDM:/\$ cd ~

arvind_2004@ARVINDM:~\$ mkdir my_project

arvind_2004@ARVINDM:~\$ cd my_project

arvind_2004@ARVINDM:~/my_project\$ git init

hint: Using 'master' as the name for the initial branch. This default branch name

hint: is subject to change. To configure the initial branch name to use in all

hint: of your new repositories, which will suppress this warning, call:

hint:

hint: git config --global init.defaultBranch <name>

hint:

hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and

hint: 'development'. The just-created branch can be renamed via this command:

hint:

hint: git branch -m <name>

Initialized empty Git repository in /home/arvind_2004/my_project/.git/

arvind_2004@ARVINDM:~/my_project\$ git add pom1.yml

fatal: pathspec 'pom1.yml' did not match any files

arvind_2004@ARVINDM:~/my_project\$ ls -l pom1.yml

ls: cannot access 'pom1.yml': No such file or directory

arvind_2004@ARVINDM:~/my_project\$ cd /

```
arvind_2004@ARVINDM:/$ cat pom1.yml
```

```
cat: pom1.yml: No such file or directory
```

```
arvind_2004@ARVINDM:/$ cat pod1.yml
```

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  name: my-pod2
```

```
spec:
```

```
  containers:
```

```
    - name: arvindm2004/webapp1
```

```
      image: arvindm2004/webapp1:tagname
```

```
  ports:
```

```
    - containerPort: 8084
```

```
arvind_2004@ARVINDM:/$ cd ~/my_project
```

```
arvind_2004@ARVINDM:~/my_project$ sudo nano pod2.yml
```

```
arvind_2004@ARVINDM:~/my_project$ git init
```

```
Reinitialized existing Git repository in /home/arvind_2004/my_project/.git/
```

```
arvind_2004@ARVINDM:~/my_project$ git add .
```

```
arvind_2004@ARVINDM:~/my_project$ git commit -m "pod"
```

```
[master (root-commit) 9e611cd] pod
```

```
1 file changed, 10 insertions(+)
```

```
create mode 100644 pod2.yml
```

```
arvind_2004@ARVINDM:~/my_project$ git remote add origin  
https://github.com/Arvind22cse/devops_tasks.git
```

```
arvind_2004@ARVINDM:~/my_project$ git checkout day4
```

```
error: pathspec 'day4' did not match any file(s) known to git
```

```
arvind_2004@ARVINDM:~/my_project$ git switch day4
```

```
fatal: invalid reference: day4
```

```
arvind_2004@ARVINDM:~/my_project$ git branch day4
```

```
arvind_2004@ARVINDM:~/my_project$ git checkout day4
```

```
Switched to branch 'day4'
```

```
arvind_2004@ARVINDM:~/my_project$ git push origin day4
```

```
Username for 'https://github.com': Arvind22cse
```

```
Password for 'https://Arvind22cse@github.com':
```

```
To https://github.com/Arvind22cse/devops_tasks.git
```

```
! [rejected]    day4 -> day4 (fetch first)
```

```
error: failed to push some refs to 'https://github.com/Arvind22cse/devops_tasks.git'
```

```
hint: Updates were rejected because the remote contains work that you do not
```

```
hint: have locally. This is usually caused by another repository pushing to
```

```
hint: the same ref. If you want to integrate the remote changes, use
```

```
hint: 'git pull' before pushing again.
```

```
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

```
arvind_2004@ARVINDM:~/my_project$ git pull
```

```
remote: Enumerating objects: 29, done.
```

```
remote: Counting objects: 100% (29/29), done.
```

```
remote: Compressing objects: 100% (27/27), done.
```

```
remote: Total 29 (delta 0), reused 29 (delta 0), pack-reused 0 (from 0)
```

```
Unpacking objects: 100% (29/29), 3.83 MiB | 2.70 MiB/s, done.
```

```
From https://github.com/Arvind22cse/devops_tasks
```

```
* [new branch]    day1    -> origin/day1
```

```
* [new branch]    day2    -> origin/day2
```

```
* [new branch]    day3    -> origin/day3
```

```
* [new branch]    day4    -> origin/day4
```

```
There is no tracking information for the current branch.
```

```
Please specify which branch you want to merge with.
```

```
See git-pull(1) for details.
```

```
git pull <remote> <branch>
```

If you wish to set tracking information for this branch you can do so with:

```
git branch --set-upstream-to=origin/<branch> day4
```

```
arvind_2004@ARVINDM:~/my_project$ git add .
```

```
arvind_2004@ARVINDM:~/my_project$ git commit -m "pod"
```

On branch day4

nothing to commit, working tree clean

```
arvind_2004@ARVINDM:~/my_project$ git push origin day4
```

Username for 'https://github.com': Arvind22cse

Password for 'https://Arvind22cse@github.com':

To https://github.com/Arvind22cse/devops_tasks.git

```
! [rejected]    day4 -> day4 (non-fast-forward)
```

error: failed to push some refs to 'https://github.com/Arvind22cse/devops_tasks.git'

hint: Updates were rejected because the tip of your current branch is behind

hint: its remote counterpart. If you want to integrate the remote changes,

hint: use 'git pull' before pushing again.

hint: See the 'Note about fast-forwards' in 'git push --help' for details.

```
arvind_2004@ARVINDM:~/my_project$ git commit -m "pod"
```

On branch day4

nothing to commit, working tree clean

```
arvind_2004@ARVINDM:~/my_project$ git push origin day4
```

Username for 'https://github.com': ^C

```
arvind_2004@ARVINDM:~/my_project$ git pull --rebase origin day4
```

From https://github.com/Arvind22cse/devops_tasks

* branch day4 -> FETCH_HEAD

Successfully rebased and updated refs/heads/day4.

arvind_2004@ARVINDM:~/my_project\$ git add .

arvind_2004@ARVINDM:~/my_project\$ git rebase --continue

fatal: No rebase in progress?

arvind_2004@ARVINDM:~/my_project\$ git push origin day4 --force-with-lease

Username for 'https://github.com': Arvind22cse

Password for 'https://Arvind22cse@github.com':

Enumerating objects: 4, done.

Counting objects: 100% (4/4), done.

Delta compression using up to 12 threads

Compressing objects: 100% (3/3), done.

Writing objects: 100% (3/3), 397 bytes | 397.00 KiB/s, done.

Total 3 (delta 0), reused 0 (delta 0), pack-reused 0

To https://github.com/Arvind22cse/devops_tasks.git

4b1990a..a515df1 day4 -> day4

arvind_2004@ARVINDM:~/my_project\$ sudo nano my-pod.yml

arvind_2004@ARVINDM:~/my_project\$ kubectl get pod

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-pod	1/1	Running	0	76m
--------	-----	---------	---	-----

my-pod2	1/1	Running	0	75m
---------	-----	---------	---	-----

mypod	1/1	Running	0	79m
-------	-----	---------	---	-----

arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-pod	1/1	Running	0	76m
--------	-----	---------	---	-----

my-pod2	1/1	Running	0	75m
---------	-----	---------	---	-----

mypod	1/1	Running	0	79m
-------	-----	---------	---	-----

arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-pod	1/1	Running	0	77m
--------	-----	---------	---	-----

my-pod2	1/1	Running	0	76m
---------	-----	---------	---	-----

mypod	1/1	Running	0	80m
-------	-----	---------	---	-----

arvind_2004@ARVINDM:~/my_project\$ pod run scale

Command 'pod' not found, did you mean:

command 'od' from deb coreutils (9.4-2ubuntu2)

command 'pdd' from deb pdd (1.7-1)

command 'pom' from deb bsdgames (2.17-30)

command 'pd' from deb puredata-core (0.54.1+ds-3)

command 'wod' from deb sope-bin (5.9.0-1)

command 'god' from deb ruby-god (0.13.7-3build5)

command 'psd' from deb profile-sync-daemon (6.50-1)

command 'poa' from deb poa (2.0+20060928-8)

command 'pox' from deb python3-pox (0.3.3-1)

command 'pwd' from deb coreutils (9.4-2ubuntu2)

command 'pods' from deb pmttools (2.2.0-3)

command 'pon' from deb ppp (2.4.9-1+1.1ubuntu1)

command 'mod' from deb monodoc-base (6.8.0.105+dfsg-3.5ubuntu1)

Try: sudo apt install <deb name>

arvind_2004@ARVINDM:~/my_project\$ kubectl apply -f my-pod.yml

pod/my-app1 created

arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-app1	0/1	ErrImagePull	0	13s
---------	-----	--------------	---	-----

my-pod	1/1	Running	0	79m
--------	-----	---------	---	-----

my-pod2	1/1	Running	0	78m
---------	-----	---------	---	-----

mypod	1/1	Running	0	82m
-------	-----	---------	---	-----

```
arvind_2004@ARVINDM:~/my_project$ sudo nano rs-test.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f rs-test.yml
```

```
error: error validating "rs-test.yml": error validating data: apiVersion not set; if you choose to ignore these errors, turn validation off with --validate=false
```

```
arvind_2004@ARVINDM:~/my_project$ sudo nano rs-test.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f rs-test.yml
```

```
error: error validating "rs-test.yml": error validating data: apiVersion not set; if you choose to ignore these errors, turn validation off with --validate=false
```

```
arvind_2004@ARVINDM:~/my_project$ sudo nano rs-test.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f rs-test.yml
```

```
replicaset.apps/my-rs created
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
my-app1	0/1	ImagePullBackOff	0	11m
my-pod	1/1	Running	0	90m
my-pod2	1/1	Running	0	89m
my-rs-7xjf6	0/1	ContainerCreating	0	8s
my-rs-fd6z6	0/1	ContainerCreating	0	8s
my-rs-h8klx	0/1	ErrImagePull	0	8s
my-rs-wcqzw	0/1	ContainerCreating	0	8s
mypod	1/1	Running	0	93m

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
my-app1	0/1	ImagePullBackOff	0	11m
my-pod	1/1	Running	0	90m
my-pod2	1/1	Running	0	89m
my-rs-7xjf6	0/1	ErrImagePull	0	11s
my-rs-fd6z6	0/1	ContainerCreating	0	11s
my-rs-h8klx	0/1	ErrImagePull	0	11s

my-rs-wcqzw 0/1 ContainerCreating 0 11s

mypod 1/1 Running 0 93m

arvind_2004@ARVINDM:~/my_project\$ kubectl delete pod my-rs-7xjf6

pod "my-rs-7xjf6" deleted

arvind_2004@ARVINDM:~/my_project\$ kubectl get pod

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-app1	0/1	ImagePullBackOff	0	12m
---------	-----	------------------	---	-----

my-pod	1/1	Running	0	91m
--------	-----	---------	---	-----

my-pod2	1/1	Running	0	90m
---------	-----	---------	---	-----

my-rs-ctmbx	0/1	ContainerCreating	0	3s
-------------	-----	-------------------	---	----

my-rs-fd6z6	0/1	ImagePullBackOff	0	66s
-------------	-----	------------------	---	-----

my-rs-h8klx	0/1	ImagePullBackOff	0	66s
-------------	-----	------------------	---	-----

my-rs-wcqzw	0/1	ErrImagePull	0	66s
-------------	-----	--------------	---	-----

mypod	1/1	Running	0	94m
-------	-----	---------	---	-----

arvind_2004@ARVINDM:~/my_project\$ kubectl exec -it my-rs-ctmbx -- /bin/bash

error: Internal error occurred: unable to upgrade connection: container not found ("my-app")

arvind_2004@ARVINDM:~/my_project\$ kubectl describe my-rs-ctmbx

error: the server doesn't have a resource type "my-rs-ctmbx"

arvind_2004@ARVINDM:~/my_project\$ kubectl describe my-rs-fd6z6

error: the server doesn't have a resource type "my-rs-fd6z6"

arvind_2004@ARVINDM:~/my_project\$ kubectl get pod

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-app1	0/1	ImagePullBackOff	0	138m
---------	-----	------------------	---	------

my-pod	1/1	Running	0	3h37m
--------	-----	---------	---	-------

my-pod2	1/1	Running	0	3h36m
---------	-----	---------	---	-------

my-rs-ctmbx	0/1	ImagePullBackOff	0	126m
-------------	-----	------------------	---	------

my-rs-fd6z6	0/1	ImagePullBackOff	0	127m
-------------	-----	------------------	---	------

my-rs-h8klx 0/1 ImagePullBackOff 0 127m

my-rs-wcqzw 0/1 ImagePullBackOff 0 127m

mypod 1/1 Running 0 3h40m

arvind_2004@ARVINDM:~/my_project\$ sudo nano deployment.yml

[sudo] password for arvind_2004:

arvind_2004@ARVINDM:~/my_project\$

arvind_2004@ARVINDM:~/my_project\$ kubectl create deployment webnginx2 --
image=nginx:latest --replicas=1

deployment.apps/webnginx2 created

arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
my-app1	0/1	ImagePullBackOff	0	145m
my-pod	1/1	Running	0	3h44m
my-pod2	1/1	Running	0	3h43m
my-rs-ctmbx	0/1	ImagePullBackOff	0	133m
my-rs-fd6z6	0/1	ImagePullBackOff	0	134m
my-rs-h8klx	0/1	ImagePullBackOff	0	134m
my-rs-wcqzw	0/1	ImagePullBackOff	0	134m
mypod	1/1	Running	0	3h47m
webnginx2-85b94f8d56-vgh6b	1/1	Running	0	13s

arvind_2004@ARVINDM:~/my_project\$ kubectl get rs

NAME	DESIRED	CURRENT	READY	AGE
my-rs	4	4	0	134m
webnginx2-85b94f8d56	1	1	1	46s

arvind_2004@ARVINDM:~/my_project\$ kubectl kubectl delete all -all

error: unknown command "kubectl" for "kubectl"

arvind_2004@ARVINDM:~/my_project\$ kubect delete all -all

Command 'kubect' not found, did you mean:

command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.3)

Try: sudo apt install <deb name>

```
arvind_2004@ARVINDM:~/my_project$ kubectl delete all -all
```

error: unknown shorthand flag: 'a' in -all

See 'kubectl delete --help' for usage.

```
arvind_2004@ARVINDM:~/my_project$ kubectl delete all --all
```

pod "my-app1" deleted

pod "my-pod" deleted

pod "my-pod2" deleted

pod "my-rs-ctmbx" deleted

pod "my-rs-fd6z6" deleted

pod "my-rs-h8klx" deleted

pod "my-rs-wcqzw" deleted

pod "mypod" deleted

pod "webnginx2-85b94f8d56-vgh6b" deleted

service "kubernetes" deleted

deployment.apps "webnginx2" deleted

replicaset.apps "my-rs" deleted

replicaset.apps "webnginx2-85b94f8d56" deleted

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-rs-kdzk7	0/1	Terminating	0	26s
-------------	-----	-------------	---	-----

my-rs-nnwdc	0/1	Terminating	0	25s
-------------	-----	-------------	---	-----

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

\Error from server (BadRequest): error when creating "deployment.yml": Deployment in version "v1" cannot be handled as a Deployment: json: cannot unmarshal string into Go struct field Container.spec.template.spec.containers.ports of type v1.ContainerPort

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

Error from server (BadRequest): error when creating "deployment.yml": Deployment in version "v1" cannot be handled as a Deployment: json: cannot unmarshal string into Go struct field Container.spec.template.spec.containers.ports of type v1.ContainerPort

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

deployment.apps/my-deployment created

```
arvind_2004@ARVINDM:~/my_project$ kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
------	-------	------------	-----------	-----

my-deployment	0/3	3	0	8s
---------------	-----	---	---	----

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-deployment-f47778678-czpqr	0/1	ErrImagePull	0	18s
-------------------------------	-----	--------------	---	-----

my-deployment-f47778678-d5bxg	0/1	ImagePullBackOff	0	18s
-------------------------------	-----	------------------	---	-----

my-deployment-f47778678-d8f7j	0/1	ImagePullBackOff	0	18s
-------------------------------	-----	------------------	---	-----

```
arvind_2004@ARVINDM:~/my_project$ kubectl describe deployment my-deployment
```

Name: my-deployment

Namespace: default

CreationTimestamp: Thu, 20 Mar 2025 08:37:01 +0000

Labels: <none>

Annotations: deployment.kubernetes.io/revision: 1

Selector: app=my-app

Replicas: 3 desired | 3 updated | 3 total | 0 available | 3 unavailable

StrategyType: RollingUpdate

MinReadySeconds: 0

RollingUpdateStrategy: 25% max unavailable, 25% max surge

Pod Template:

Labels: app=my-app

Containers:

my-app:

Image: arvindm2004/webapp1:tagname

Port: 8089/TCP

Host Port: 0/TCP

Environment: <none>

Mounts: <none>

Volumes: <none>

Node-Selectors: <none>

Tolerations: <none>

Conditions:

Type	Status	Reason
------	--------	--------

----	-----	-----
------	-------	-------

Available	False	MinimumReplicasUnavailable
-----------	-------	----------------------------

Progressing	True	ReplicaSetUpdated
-------------	------	-------------------

OldReplicaSets: <none>

NewReplicaSet: my-deployment-f47778678 (3/3 replicas created)

Events:

Type	Reason	Age	From	Message
------	--------	-----	------	---------

----	-----	----	----	-----
------	-------	------	------	-------

Normal	ScalingReplicaSet	26s	deployment-controller	Scaled up replica set my-deployment-f47778678 from 0 to 3
--------	-------------------	-----	-----------------------	---

arvind_2004@ARVINDM:~/my_project\$ sudo nano deployment.yml

arvind_2004@ARVINDM:~/my_project\$ kubectl create deployment webnginx2 --image=nginx:latest --replicas=1

deployment.apps/webnginx2 created

arvind_2004@ARVINDM:~/my_project\$ kubectl replace -f deployment.yml

deployment.apps/my-deployment replaced

arvind_2004@ARVINDM:~/my_project\$ kubectl get pod

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-deployment-f47778678-5qspr	0/1	ContainerCreating	0	12s
-------------------------------	-----	-------------------	---	-----

```

my-deployment-f47778678-c8d5m 0/1 ErrImagePull 0 12s
my-deployment-f47778678-czpqr 0/1 ImagePullBackOff 0 4m32s
my-deployment-f47778678-d5bxg 0/1 ImagePullBackOff 0 4m32s
my-deployment-f47778678-d8f7j 0/1 ImagePullBackOff 0 4m32s
my-deployment-f47778678-f82gp 0/1 ContainerCreating 0 12s
my-deployment-f47778678-fml6f 0/1 ErrImagePull 0 12s
my-deployment-f47778678-zrh84 0/1 ContainerCreating 0 12s
webnginx2-85b94f8d56-gsz4q 1/1 Running 0 72s
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy my-deploy --replicas=2
error: no objects passed to scale deployments.apps "my-deploy" not found
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy mydeploy --replicas=2
error: no objects passed to scale deployments.apps "mydeploy" not found
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy my-deploy --replicas=2
error: no objects passed to scale deployments.apps "my-deploy" not found
arvind_2004@ARVINDM:~/my_project$ kubectl scale deploy my-deployment --replicas=2
deployment.apps/my-deployment scaled
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME READY STATUS RESTARTS AGE
my-deployment-f47778678-czpqr 0/1 ImagePullBackOff 0 10m
my-deployment-f47778678-d8f7j 0/1 ImagePullBackOff 0 10m
webnginx2-85b94f8d56-gsz4q 1/1 Running 0 7m13s
arvind_2004@ARVINDM:~/my_project$ kubectl delete webnginx2-85b94f8d56-gsz4q
error: the server doesn't have a resource type "webnginx2-85b94f8d56-gsz4q"
arvind_2004@ARVINDM:~/my_project$ kubectl delete pod webnginx2-85b94f8d56-gsz4q
pod "webnginx2-85b94f8d56-gsz4q" deleted
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
NAME READY STATUS RESTARTS AGE
my-deployment-f47778678-czpqr 0/1 ImagePullBackOff 0 11m

```

```
my-deployment-f47778678-d8f7j 0/1 ImagePullBackOff 0 11m
```

```
webnginx2-85b94f8d56-cv xv2 1/1 Running 0 6s
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-deployment-f47778678-czpqr	0/1	ImagePullBackOff	0	11m
-------------------------------	-----	------------------	---	-----

my-deployment-f47778678-d8f7j	0/1	ErrImagePull	0	11m
-------------------------------	-----	--------------	---	-----

webnginx2-85b94f8d56-cv xv2	1/1	Running	0	10s
-----------------------------	-----	---------	---	-----

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

my-deployment-f47778678-czpqr	0/1	ImagePullBackOff	0	12m
-------------------------------	-----	------------------	---	-----

my-deployment-f47778678-d8f7j	0/1	ImagePullBackOff	0	12m
-------------------------------	-----	------------------	---	-----

webnginx2-85b94f8d56-cv xv2	1/1	Running	0	29s
-----------------------------	-----	---------	---	-----

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl create apply -f deployment.yml
```

```
error: Unexpected args: [apply]
```

```
See 'kubectl create -h' for help and examples
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl create apply -f deployment.yml
```

```
error: Unexpected args: [apply]
```

```
See 'kubectl create -h' for help and examples
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

```
deployment.apps/my-deploy created
```

```
Error from server (BadRequest): error when creating "deployment.yml": Service in version "v1" cannot be handled as a Service: json: cannot unmarshal string into Go struct field ServiceSpec.spec.selector of type map[string]string
```

```
arvind_2004@ARVINDM:~/my_project$ cat deployment.yml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: my-deploy
```

labels:

name: my-deploy

spec:

replicas: 1

selector:

matchLabels:

apptype: web-backend

strategy:

type: RollingUpdate

template:

metadata:

labels:

apptype: web-backend

spec:

containers:

- name: my-app

image: arvindm2004/webapp1:tagname

ports:

- containerPort: 7070

apiVersion: v1

kind: Service

metadata:

name: my-service

labels:

app: my-service

type: backend-app

spec:

type: NodePort

ports:

- targetPort: 7070

port: 7070

nodePort: 30002

selector:

apptype: web-backend

arvind_2004@ARVINDM:~/my_project\$ sudo nano deployment.yml

[sudo] password for arvind_2004:

arvind_2004@ARVINDM:~/my_project\$ kubectl apply -f deployment.yml

service/my-service created

arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
my-deploy-5df65497db-xfhlq	0/1	ErrImagePull	0	3m40s
my-deployment-f47778678-czpqr	0/1	ImagePullBackOff	0	41m
my-deployment-f47778678-d8f7j	0/1	ImagePullBackOff	0	41m
webnginx2-85b94f8d56-cvxx2	1/1	Running	0	29m

arvind_2004@ARVINDM:~/my_project\$ kubectl get services

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	44m
my-service	NodePort	10.106.84.41	<none>	7070:30002/TCP	31s

arvind_2004@ARVINDM:~/my_project\$ sudo nano deployment.yml

[sudo] password for arvind_2004:

arvind_2004@ARVINDM:~/my_project\$ kubectl apply -f deployment.yml

deployment.apps/my-deploy configured

service/my-service unchanged

arvind_2004@ARVINDM:~/my_project\$ kubectl replace deployment.yml

error: must specify one of -f and -k

arvind_2004@ARVINDM:~/my_project\$ kubectl replace -f deployment.yml

deployment.apps/my-deploy replaced

service/my-service replaced

arvind_2004@ARVINDM:~/my_project\$ minikube service my-service

-----	-----	-----	-----
NAMESPACE	NAME	TARGET PORT	URL
-----	-----	-----	-----
default	my-service	7070	http://192.168.49.2:30002
-----	-----	-----	-----

✗ Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found

🐱 If the above advice does not help, please let us know:	
📄 https://github.com/kubernetes/minikube/issues/new/choose	
Please run `minikube logs --file=logs.txt` and attach logs.txt to the GitHub issue.	
Please also attach the following file to the GitHub issue:	
- /tmp/minikube_service_8aaa71feca0d33a15dc0b5f1d3de1c519de02edd_0.log	

arvind_2004@ARVINDM:~/my_project\$ curl http://192.168.49.2:30002

curl: (7) Failed to connect to 192.168.49.2 port 30002 after 1 ms: Couldn't connect to server

arvind_2004@ARVINDM:~/my_project\$ ping http://192.168.49.2:30002

ping: http://192.168.49.2:30002: Name or service not known

arvind_2004@ARVINDM:~/my_project\$ ping http://192.168.49.2:30002/my-app/

ping: http://192.168.49.2:30002/my-app/: Name or service not known

arvind_2004@ARVINDM:~/my_project\$ sudo chmod 666 /var/run/docker.sock

[sudo] password for arvind_2004:

arvind_2004@ARVINDM:~/my_project\$ kubectl delete all --all

pod "my-deploy-5df65497db-mv5jq" deleted

pod "my-deploy-5df65497db-n4ln2" deleted

pod "my-deploy-5df65497db-tt8n6" deleted

pod "my-deploy-5df65497db-xfhlq" deleted

pod "my-deployment-f47778678-czpqr" deleted

pod "my-deployment-f47778678-d8f7j" deleted

pod "webnginx2-85b94f8d56-cvxx2" deleted

service "kubernetes" deleted

service "my-service" deleted

deployment.apps "my-deploy" deleted

deployment.apps "my-deployment" deleted

deployment.apps "webnginx2" deleted

replicaset.apps "webnginx2-85b94f8d56" deleted

arvind_2004@ARVINDM:~/my_project\$ sudo nano deployment.yml

arvind_2004@ARVINDM:~/my_project\$ sudo nano deployment.yml

arvind_2004@ARVINDM:~/my_project\$ kubectl apply -f deployment.yml

deployment.apps/my-deploy created

service/my-service created

arvind_2004@ARVINDM:~/my_project\$ minikube service myservice.yml

✗ Exiting due to SVC_NOT_FOUND: Service 'myservice.yml' was not found in 'default' namespace.

You may select another namespace by using 'minikube service myservice.yml -n <namespace>'. Or list out all the services using 'minikube service list'

```
arvind_2004@ARVINDM:~/my_project$ minikube service my-service
```

-----	-----	-----	-----
NAMESPACE	NAME	TARGET PORT	URL
-----	-----	-----	-----
default	my-service	9000	http://192.168.49.2:30002
-----	-----	-----	-----

✗ Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found

🐱 If the above advice does not help, please let us know:	
📄 https://github.com/kubernetes/minikube/issues/new/choose	
Please run `minikube logs --file=logs.txt` and attach logs.txt to the GitHub issue.	
Please also attach the following file to the GitHub issue:	
- /tmp/minikube_service_8aaa71feca0d33a15dc0b5f1d3de1c519de02edd_0.log	

```
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30002
```

```
curl: (7) Failed to connect to 192.168.49.2 port 30002 after 0 ms: Couldn't connect to server
```

```
arvind_2004@ARVINDM:~/my_project$ cat deployment.yml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: my-deploy
```

```
  labels:
```

```
    name: my-deploy
```

```
spec:
```

```
  replicas: 4
```

```
  selector:
```

```
    matchLabels:
```

```
      apptype: web-backend
```

```
  strategy:
```

```
    type: RollingUpdate
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        apptype: web-backend
```

```
    spec:
```

```
      containers:
```

```
        - name: my-app
```

```
          image: arvindm2004/webapp1:tagname
```

```
          ports:
```

```
            - containerPort: 9000
```

```
---
```

apiVersion: v1

kind: Service

metadata:

name: my-service

labels:

app: my-service

type: backend-app

spec:

type: NodePort

ports:

- targetPort: 9000

port: 9000

nodePort: 30002

selector:

apptype: web-backend

arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
my-deploy-84f4966d79-64z7p	0/1	ImagePullBackOff	0	4m41s
my-deploy-84f4966d79-7f92t	0/1	ImagePullBackOff	0	4m41s
my-deploy-84f4966d79-bt29f	0/1	ImagePullBackOff	0	4m41s
my-deploy-84f4966d79-n4ts5	0/1	ImagePullBackOff	0	4m41s

arvind_2004@ARVINDM:~/my_project\$ kubectl describe deployment my-deploy

Name: my-deploy

Namespace: default

CreationTimestamp: Thu, 20 Mar 2025 10:07:31 +0000

Labels: name=my-deploy

Annotations: deployment.kubernetes.io/revision: 1

Selector: apptype=web-backend

Replicas: 4 desired | 4 updated | 4 total | 0 available | 4 unavailable

StrategyType: RollingUpdate

MinReadySeconds: 0

RollingUpdateStrategy: 25% max unavailable, 25% max surge

Pod Template:

Labels: apptype=web-backend

Containers:

my-app:

Image: arvindm2004/webapp1:tagname

Port: 9000/TCP

Host Port: 0/TCP

Environment: <none>

Mounts: <none>

Volumes: <none>

Node-Selectors: <none>

Tolerations: <none>

Conditions:

Type	Status	Reason
------	--------	--------

----	-----	-----
------	-------	-------

Available	False	MinimumReplicasUnavailable
-----------	-------	----------------------------

Progressing	True	ReplicaSetUpdated
-------------	------	-------------------

OldReplicaSets: <none>

NewReplicaSet: my-deploy-84f4966d79 (4/4 replicas created)

Events:

Type	Reason	Age	From	Message
------	--------	-----	------	---------

----	-----	----	----	-----
------	-------	------	------	-------

Normal	ScalingReplicaSet	4m54s	deployment-controller	Scaled up replica set my-deploy-84f4966d79 from 0 to 4
--------	-------------------	-------	-----------------------	--

```
arvind_2004@ARVINDM:~/my_project$ kubectl port-forward service/my-service 9000:9000
```

```
error: unable to forward port because pod is not running. Current status=Pending
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl delete all --all
```

```
pod "my-deploy-84f4966d79-64z7p" deleted
```

```
pod "my-deploy-84f4966d79-7f92t" deleted
```

```
pod "my-deploy-84f4966d79-bt29f" deleted
```

```
pod "my-deploy-84f4966d79-n4ts5" deleted
```

```
service "kubernetes" deleted
```

```
service "my-service" deleted
```

```
deployment.apps "my-deploy" deleted
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl delete pods --all
```

```
pod "my-deploy-84f4966d79-52xlz" deleted
```

```
pod "my-deploy-84f4966d79-rptt5" deleted
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

```
deployment.apps/my-deploy created
```

```
service/my-service created
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
my-deploy-84f4966d79-qhtb2	0/1	ContainerCreating	0	9s
my-deploy-84f4966d79-tg7nl	0/1	ContainerCreating	0	9s
my-deploy-84f4966d79-zflkr	0/1	ErrImagePull	0	9s
my-deploy-84f4966d79-zgf99	0/1	ContainerCreating	0	9s

```
arvind_2004@ARVINDM:~/my_project$ kubectl port-forward service/my-service 9000:9000
```

```
error: unable to forward port because pod is not running. Current status=Pending
```

```
arvind_2004@ARVINDM:~/my_project$ docker pull arvindm2004/webapp1:tagname
```

```
Error response from daemon: manifest for arvindm2004/webapp1:tagname not found:  
manifest unknown: manifest unknown
```

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

```
deployment.apps/my-deploy configured
```

```
error: error parsing deployment.yml: error converting YAML to JSON: yaml: line 15: mapping values are not allowed in this context
```

```
arvind_2004@ARVINDM:~/my_project$ minikube service my-service
```

-----	-----	-----	-----
NAMESPACE	NAME	TARGET PORT	URL
-----	-----	-----	-----
default	my-service	9000	http://192.168.49.2:30002
-----	-----	-----	-----

✗ Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found

🐱 If the above advice does not help, please let us know:	
📄 https://github.com/kubernetes/minikube/issues/new/choose	
Please run `minikube logs --file=logs.txt` and attach logs.txt to the GitHub issue.	
Please also attach the following file to the GitHub issue:	
- /tmp/minikube_service_8aaa71feca0d33a15dc0b5f1d3de1c519de02edd_0.log	

```
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30002
```

curl: (7) Failed to connect to 192.168.49.2 port 30002 after 0 ms: Couldn't connect to server

arvind_2004@ARVINDM:~/my_project\$ kubectl replace deployment.yml

error: must specify one of -f and -k

arvind_2004@ARVINDM:~/my_project\$ kubectl replace -f deployment.yml

deployment.apps/my-deploy replaced

error: error parsing deployment.yml: error converting YAML to JSON: yaml: line 15: mapping values are not allowed in this context

arvind_2004@ARVINDM:~/my_project\$ cat deployment.yml

apiVersion: apps/v1

kind: Deployment

metadata:

name: my-deploy

labels:

name: my-deploy

spec:

replicas: 1

selector:

matchLabels:

apptype: web-backend

strategy:

type: RollingUpdate

template:

metadata:

labels:

apptype: web-backend

spec:

containers:

- name: my-app

image: arvindm2004/webapp1:tagname

ports:

- containerPort: 9000

apiVersion: v1

kind: Service

metadata:

name: my-service

labels:

app: my-service

spec:

type: NodePort

ports:

- port: 9000

targetPort: 8080

nodePort: 30002

selector:

apptype: web-backendapiVersion: apps/v1

kind: Deployment

metadata:

name: my-deploy

labels:

name: my-deploy

spec:

replicas: 1

selector:

matchLabels:

apptype: web-backend

strategy:

type: RollingUpdate

template:

metadata:

labels:

apptype: web-backend

spec:

containers:

- name: my-app

image: arvindm2004/webapp1:tagname

ports:

- containerPort: 9000

apiVersion: v1

kind: Service

metadata:

name: my-service

labels:

app: my-service

spec:

type: NodePort

ports:

- port: 9000

targetPort: 8080

nodePort: 30002

selector:

apptype: web-backend

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

Warning: resource deployments/my-deploy is missing the kubectl.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.

deployment.apps/my-deploy configured

service/my-service configured

```
arvind_2004@ARVINDM:~/my_project$ minikube service my-service
```

-----	-----	-----	-----
NAMESPACE	NAME	TARGET PORT	URL
-----	-----	-----	-----
default	my-service	9000	http://192.168.49.2:30002
-----	-----	-----	-----

✖ Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found

🐱 If the above advice does not help, please let us know:	
📄 https://github.com/kubernetes/minikube/issues/new/choose	
Please run `minikube logs --file=logs.txt` and attach logs.txt to the GitHub issue.	
Please also attach the following file to the GitHub issue:	

```
| - /tmp/minikube_service_8aaa71feca0d33a15dc0b5f1d3de1c519de02edd_0.log
```

```
|
```

```
|
```

```
|
```

```
arvind_2004@ARVINDM:~/my_project$ minikube log
```

```
Error: unknown command "log" for "minikube"
```

```
Did you mean this?
```

```
logs
```

```
Run 'minikube --help' for usage.
```

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
```

```
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
```

```
deployment.apps/my-deploy configured
```

```
service/my-service unchanged
```

```
arvind_2004@ARVINDM:~/my_project$ minikube service my-service
```

```
|-----|-----|-----|-----|
```

```
| NAMESPACE | NAME   | TARGET PORT |    URL    |
```

```
|-----|-----|-----|-----|
```

```
| default   | my-service |    9000    | http://192.168.49.2:30002 |
```

```
|-----|-----|-----|-----|
```

```
✗ Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found
```

```

|                                     |
| 🐱 If the above advice does not help, please let us know: |
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| - /tmp/minikube_service_8aaa71feca0d33a15dc0b5f1d3de1c519de02edd_0.log
|
|                                     |
|_____
|_____

```

```

arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
arvind_2004@ARVINDM:~/my_project$ kubectl delete all --all
pod "my-deploy-84f4966d79-qhtb2" deleted
service "kubernetes" deleted
service "my-service" deleted
deployment.apps "my-deploy" deleted
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
deployment.apps/my-deploy created
service/my-service created
arvind_2004@ARVINDM:~/my_project$ minikube service my-service
|-----|-----|-----|-----|
| NAMESPACE | NAME   | TARGET PORT |    URL    |
|-----|-----|-----|-----|
| default   | my-service | 9001 | http://192.168.49.2:30002 |
|-----|-----|-----|-----|

```

✖ Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found

```
|
|
|
| 🐱 If the above advice does not help, please let us know:
| 👉 https://github.com/kubernetes/minikube/issues/new/choose
|
|
| Please run `minikube logs --file=logs.txt` and attach logs.txt to the GitHub issue.
| Please also attach the following file to the GitHub issue:
| - /tmp/minikube_service_8aaa71feca0d33a15dc0b5f1d3de1c519de02edd_0.log
|
|
|
```

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
arvind_2004@ARVINDM:~/my_project$ kubectl delete all --all
pod "my-deploy-6b478cb7cb-4fhvv" deleted
service "kubernetes" deleted
service "my-service" deleted
deployment.apps "my-deploy" deleted
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
deployment.apps/my-deploy created
service/my-service created
arvind_2004@ARVINDM:~/my_project$ minikube service my-service
|-----|-----|-----|-----|
| NAMESPACE | NAME   | TARGET PORT |    URL    |
|
```

default	my-service	9001	http://192.168.49.2:30005
---------	------------	------	---------------------------

 Starting tunnel for service my-service.

NAMESPACE	NAME	TARGET PORT	URL
default	my-service		http://127.0.0.1:36227

 Opening service default/my-service in default browser...

 <http://127.0.0.1:36227>

 Because you are using a Docker driver on linux, the terminal needs to be open to run it.

 Stopping tunnel for service my-service.

arvind_2004@ARVINDM:~/my_project\$ curl http://192.168.49.2:30005

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-
size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-
color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not
Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b>
The origin server did not find a current representation for the target resource or is not
willing to disclose that one exists.</p><hr class="line" /><h3>Apache
Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:~/my_project$ curl
http://192.168.49.2:30005/my-app/
```

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-
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Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b>
The origin server did not find a current representation for the target resource or is not
willing to disclose that one exists.</p><hr class="line" /><h3>Apache
Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:kubect get pod
```

Command 'kubect' not found, did you mean:

command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.3)

Try: sudo apt install <deb name>

arvind_2004@ARVINDM:~/my_project\$ kubectl get pod

NAME	READY	STATUS	RESTARTS	AGE
my-deploy-6b478cb7cb-t5lkg	1/1	Running	0	3m38s

arvind_2004@ARVINDM:~/my_project\$ kubectl exec -it my-deploy-my-deploy-6b478cb7cb-t5lkg -- /bin/bash

Error from server (NotFound): pods "my-deploy-my-deploy-6b478cb7cb-t5lkg" not found

arvind_2004@ARVINDM:~/my_project\$ minikube service my-service

NAMESPACE	NAME	TARGET PORT	URL
default	my-service	9001	http://192.168.49.2:30005

 Starting tunnel for service my-service.

NAMESPACE	NAME	TARGET PORT	URL
default	my-service		http://127.0.0.1:35123

 Opening service default/my-service in default browser...

 http://127.0.0.1:35123

 Because you are using a Docker driver on linux, the terminal needs to be open to run it.

^C  Stopping tunnel for service my-service.

arvind_2004@ARVINDM:~/my_project\$ curl http://127.0.0.1:35123

curl: (7) Failed to connect to 127.0.0.1 port 35123 after 0 ms: Couldn't connect to server

arvind_2004@ARVINDM:~/my_project\$ curl http://192.168.49.2:30005

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-
size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-
color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not
Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b>
The origin server did not find a current representation for the target resource or is not
willing to disclose that one exists.</p><hr class="line" /><h3>Apache
Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:curl
http://192.168.49.2:30005/my-app/
```

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-
size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-
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Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b>
The origin server did not find a current representation for the target resource or is not
willing to disclose that one exists.</p><hr class="line" /><h3>Apache
Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:curl
http://192.168.49.2:30005/my-service/
```

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-
size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-
color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not
Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b>
The origin server did not find a current representation for the target resource or is not
willing to disclose that one exists.</p><hr class="line" /><h3>Apache
Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:~/my_project$ curl
http://192.168.49.2:30005/my-app
```

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style
type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-
size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-
color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not
Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b>
The origin server did not find a current representation for the target resource or is not
willing to disclose that one exists.</p><hr class="line" /><h3>Apache
```

Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:~/my_project\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
my-deploy-6b478cb7cb-t5lkg	1/1	Running	0	6m50s

arvind_2004@ARVINDM:~/my_project\$ curl http://192.168.49.2:30005/my-deploy

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b> The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /><h3>Apache Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:curl http://192.168.49.2:30005/my-deploy/
```

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b> The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /><h3>Apache Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-deploy/
```

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b> The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /><h3>Apache Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:curl http://192.168.49.2:30005/maven-web-app/
```

```
<!doctype html><html lang="en"><head><title>HTTP Status 404 – Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b
```

```
{color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 – Not Found</h1><hr class="line" /><p><b>Type</b> Status Report</p><p><b>Description</b> The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /><h3>Apache Tomcat/9.0.102</h3></body></html>arvind_2004@ARVINDM:~/my_project$ kubectl exec -it my-deploy-6b478cb7cb-t5lkg -- /bin/bash
```

```
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat# ls
```

```
bin      conf      filtered-KEYS  LICENSE  native-jni-lib  README.md  RUNNING.txt
upstream-KEYS  webapps.dist
```

```
BUILDING.txt  CONTRIBUTING.md  lib      logs  NOTICE  RELEASE-NOTES  temp
webapps      work
```

```
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat# cd webapps
```

```
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat/webapps# ls
```

```
my-web  my-web.war
```

```
root@my-deploy-6b478cb7cb-t5lkg:/usr/local/tomcat/webapps# cd
```

```
root@my-deploy-6b478cb7cb-t5lkg:~# exit
```

```
exit
```

```
arvind_2004@ARVINDM:~/my_project$ ls
```

```
deployment.yml  kubctl.txt  my-pod.yml  pod2.yml  rs-test.yml
```

```
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-web/
```

```
<html>
```

```
<body>
```

```
<h2>Hello World!</h2>
```

```
</body>
```

```
</html>
```

```
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
```

```
arvind_2004@ARVINDM:~/my_project$ curl http://192.168.49.2:30005/my-web/
```

```
<html>
```

```
<body>
```

<h2>Hello World!</h2>

</body>

</html>

```
arvind_2004@ARVINDM: ~/m
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
service/my-service created
arvind_2004@ARVINDM:~/my_project$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
my-deploy-5df65497db-xfh1q          0/1     ErrImagePull       0           3m40s
my-deployment-f47778678-czpqr       0/1     ImagePullBackOff   0           41m
my-deployment-f47778678-d8f7j       0/1     ImagePullBackOff   0           41m
webnginx2-85b94f8d56-cvxxv2        1/1     Running            0           29m
arvind_2004@ARVINDM:~/my_project$ kubectl get services
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes  ClusterIP   10.96.0.1    <none>        443/TCP          44m
my-service  NodePort    10.106.84.41 <none>        7070:30002/TCP   31s
arvind_2004@ARVINDM:~/my_project$ sudo nano deployment.yml
[sudo] password for arvind_2004:
arvind_2004@ARVINDM:~/my_project$ kubectl apply -f deployment.yml
deployment.apps/my-deploy configured
service/my-service unchanged
arvind_2004@ARVINDM:~/my_project$ kubectl replace deployment.yml
error: must specify one of -f and -k
arvind_2004@ARVINDM:~/my_project$ kubectl replace -f deployment.yml
deployment.apps/my-deploy replaced
service/my-service replaced
arvind_2004@ARVINDM:~/my_project$ minikube service my-service

```

NAMESPACE	NAME	TARGET PORT	URL
default	my-service	7070	http://192.168.49.2:30002

```

X Exiting due to SVC_UNREACHABLE: service not available: no running pod for service my-service found

```